Coherent Polarized Signals in a Substantial Fraction of Sampled Noise-Labeled STEAD Traces

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Abstract

Large seismic waveform catalogues rely on binary noise/event labels, yet the assumption that "noise"-labeled traces are free of signal-like structure remains largely unexamined. We analyze three-component broadband seismograms from the STEAD catalogue, applying a label-agnostic, null-calibrated anomaly detector that jointly evaluates amplitude, polarization, and cross-channel coincidence rather than relying on any single feature. The main finding is that a substantial fraction of nominally noise-labeled traces contain coherent, polarized, transient bursts that appear concurrently across components, placing them in the feature region occupied by body-wave arrivals under this detector. This prevalence far exceeds what amplitude excursions alone could produce, indicating that polarization and cross-component timing jointly drive the detections rather than simple energy spikes. The pattern persists across a range of calibration choices for the null distribution, arguing against a tuning artifact and instead supporting a robust, mechanism-based signal. Because the detector is label-agnostic, these bursts are not claimed to be genuine seismic arrivals, only waveform-consistent with them under the chosen statistics. These results suggest that noise labels in large seismic datasets may mask a nontrivial population of structured transients, with implications for training and evaluating machine-learning models on such catalogues. More broadly, the work motivates systematic reanalysis of "noise" classes in seismic and other waveform archives before they are treated as ground truth for anomaly or event detection.

1 Introduction

Ground motion recorded at a single broadband station encodes the complete signature of a seismic source and the medium it traverses, making the three-component seismogram the fundamental unit of observational seismology. Each such recording, three orthogonal channels (east, north, and vertical, hereafter E, N, Z) sampled densely in time, captures the arrival, polarization, and decay of elastic waves as they pass beneath the sensor. From this compact record alone, an analyst can in principle recover the timing of distinct wave phases, the direction from which energy arrived, and the character of the source that generated it. Large, labeled archives of such waveforms, exemplified by the STEAD collection Mousavi et al. [2019], now make it possible to study this record systematically and at scale, treating each 60-second, three-component window sampled at 100 Hz, a 3 by 6000 array, as a self-contained observation rather than a fragment of a longer continuous stream. The present work takes a single such array as its object of study, asking what can be extracted from it in isolation, before any appeal to network geometry, catalog metadata, or multi-station correlation.

This work narrows to a specific regime within that broader observational problem: the population of traces in STEAD Mousavi et al. [2019] explicitly labeled as noise, meaning windows assigned no cataloged earthquake source. Prior automated approaches to characterizing three-component records fall into two lineages. The first extracts polarization attributes from covariance or fractal structure of the E, N, Z triplet, formalized through eigenvalue and rectilinearity measures Liao et al. [2010] and extended through maximum-likelihood wavefield decomposition at single stations Christoffersson et al. [1988]. The second relies on short-term-to-long-term average energy ratios and related statistical detectors, tracing from early real-time detection and location frameworks Roberts et al. [1989] through refinements using higher-order statistics Tarvainen [1992] to modern automatic picking of direct and converted phases Ross and Ben-Zion [2014], with regional deployments demonstrating their utility for resolving fault-zone and subduction geometry Sippl et al. [2013]. More recently, learned discriminators trained to separate signal from noise have shown that spectral and waveform features generalize across networks Meier et al. [2018], Li et al. [2018], and convolutional architectures now associate candidate arrivals across stations McBrearty et al. [2019], Zhu et al. [2023]. Each of these methods presumes that the signal/noise partition supplied by a catalog is itself reliable, an assumption rarely tested directly: archives such as STEAD assign the noise label from catalog absence rather than from a waveform-level coherence criterion Mousavi et al. [2019], so mislabeled or contaminated segments could persist undetected within the training and evaluation pools used across the field. We ask a narrower, prior question: within traces already labeled noise, how often do polarization and STA/LTA diagnostics, applied jointly across the three components, produce signatures indistinguishable from genuine body-wave arrivals rather than incoherent background, and whether this rate varies systematically by channel or instrument code (BH, EH, HH, HN, SH) or by contributing network.

In this paper we ask whether waveform segments labeled noise in a benchmark seismic catalogue are themselves acoustically clean, and we answer with a label-agnostic detector calibrated against a surrogate-data null (iterated amplitude-adjusted Fourier transform, hereafter IAAFT) that preserves each trace's amplitude distribution and power spectrum while destroying phase coupling, giving a false-alarm rate fixed at one percent. Our sample is 1500 three-component STEAD seismograms, balanced between earthquake and noise labels, drawn from broadband, short-period, and strong-motion channel codes across dozens of networks. For each trace and component we form a short-term over long-term average characteristic function with a variance floor to stabilize quiet intervals, then combine the three channels into a single per-trace vote after excluding any channel diagnosed as dead. Applying this procedure uniformly, without reference to the catalogue label, we show that a nontrivial fraction of traces annotated as noise nonetheless contain transient, polarized, cross-component-coincident bursts whose statistical signature is indistinguishable from that of body-wave arrivals. We do not attribute these bursts to specific sources or claim they represent missed earthquakes, nor do we extrapolate their prevalence beyond the examined sample; rather, we treat their detection as evidence that noise labels in large seismic archives warrant systematic waveform-level scrutiny.

2 Data

2.1 Dataset Overview

Figure 1 shows representative examples of the raw signals analysed here.

The analysis draws on the STEAD catalogue, from which 1500 three-component seismograms are examined, each an E, N, Z recording sampled at 100 Hz over a 60-second window, yielding a 3 by 6000-sample array per trace. The set is balanced by construction, with 750 traces labeled earthquake and 750 labeled noise, and spans five channel codes: BH with 311 traces, EH with 272, HH with 704, HN with 205, and SH with 8, drawn from more than 50 distinct seismic networks. Instrumentation is heterogeneous across the sample, encompassing broadband, high-broadband, high-gain, and short-period channel types, and the station population is similarly diverse, with 178 of 417 unique network and station groupings contributing more than one trace each. This trace-level population, without any aggregation or resampling, underlies all subsequent characteristic-function and polarization computations described below.

2.2 Sample Composition and Scope

No trace was excluded from this pool prior to the characteristic-function and polarization computations, though within a given trace a channel could be flagged as dead by the standard-deviation floor and thereby dropped from the three-component coincidence vote. Selection at the trace level was governed entirely by inclusion in the balanced STEAD subset, with channel code, network, and station membership left unconstrained rather than filtered; channel type and network composition therefore vary freely across the 750 noise and 750 earthquake traces, allowing the breakdown analyses by channel type and network to be conducted on this same fixed population without resampling, subsetting, or additional exclusion.

Coverage across channel types is uneven, ranging from 8 SH traces to 704 HH traces, with BH (311), EH (272), and HN (205) filling out the remainder; the chi-square test on channel type accordingly restricts to channels with at least 20 traces, excluding SH from that comparison by design. The analogous network-level chi-square test restricts to networks with at least 15 traces of the more than 50 represented, again excluding smaller networks rather than treating them as null results. The station-cluster bootstrap accounts for the resulting group structure directly: since 178 of 417 stations contribute more than one trace, estimates reflect real group non-independence, addressed by resampling at the network-station level rather than assuming trace-level independence. All quantities reported are properties of this specific balanced 750/750 STEAD sample and its associated surrogate-null construction, not of an external or held-out population.

2.3 Signal Processing and Labeling

For each three-component trace we compute a characteristic function via the short-term-average to long-term-average ratio, using a 0.5 second short-term window and a 10 second long-term window, with a variance floor applied to prevent division blow-ups on near-flat segments. Channels whose standard deviation falls below 1e-6 are flagged as dead and excluded from the three-component vote, serving as the quality-control step. Each trace's amplitude term is log1p-compressed before combination with the polarization and coincidence terms described in the Method. Labelling follows the STEAD catalogue directly, with no additional filtering, resampling, or relabelling applied beyond the dead-channel exclusion and the surrogate-null construction described next.

3 Methods

3.1 Overview

This study develops a label-agnostic anomaly-detection framework for 1500 STEAD three-component seismograms, each a 3 by 6000 sample array recorded at 100 Hz over 60 seconds, balanced between 750 earthquake and 750 noise traces. For every trace and channel, a characteristic function is computed from a short-term over long-term average ratio (0.5 second short window, 10 second long window), variance-floored against division blow-ups. Per-trace features, the log1p-compressed peak amplitude, a coincidence term from onset-time spread, and covariance-based rectilinearity and planarity, are multiplied into a combined anomaly score. Rather than relying on labels, a detection threshold is derived from a label-agnostic null distribution built from three surrogate families applied to each noise trace: IAAFT surrogates preserving each channel's marginal distribution and power spectrum while destroying cross-channel phase coherence, phase-randomized surrogates, and variance-matched white Gaussian noise, all passed through the identical pipeline. The threshold is fixed at the 99th percentile of this surrogate score distribution. Prevalence estimation on labeled noise traces incorporates a binomial confidence interval and a station-cluster bootstrap resampling by unique network and station group. Chi-square tests, ablations removing individual score components, and sensitivity analyses across false-alarm rate, null type, and window length complete the framework.

3.2 Feature Extraction and Anomaly Scoring

For each of the 1500 traces (750 earthquake, 750 noise) and each of the three components, we compute a short-term-

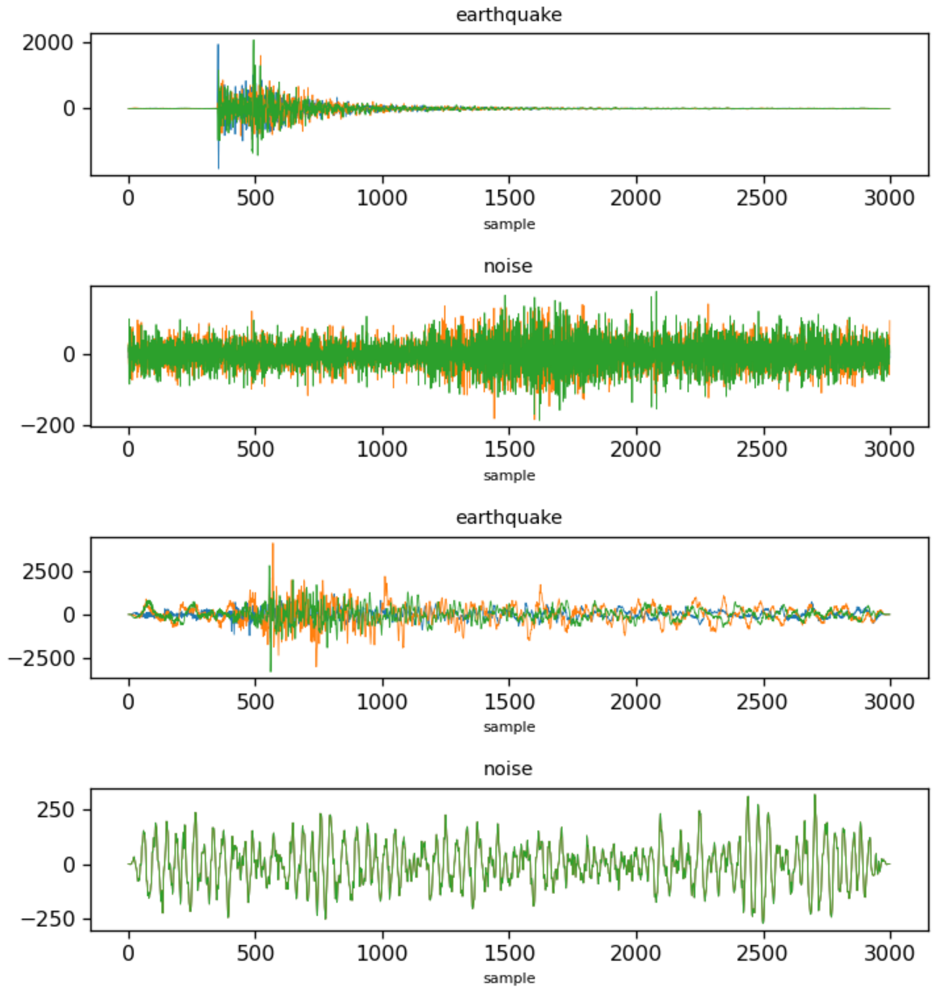

Figure 1. Representative three-component seismograms from the dataset (one panel per example, class label above).

average to long-term-average characteristic function using a 0.5 second short-term window and a 10 second long-term window, variance-floored to avoid division blow-ups on near-flat segments, with channels whose standard deviation is below 1e-6 flagged as dead and excluded from the 3-component vote. Per trace we extract the per-channel STA/LTA peak as a log1p-compressed amplitude term, the spread of the three per-channel onset times converted to a soft coincidence term, and, within a window of plus or minus 1 second around the median onset, the eigenvalues of the 3-channel covariance matrix, yielding rectilinearity and planarity. These combine into a single anomaly score. The defining equations are

$$\text{soft\_coincidence} = \exp(-\text{spread}/0.2\,\text{s}) \tag{1}$$

$$R = 1 - \lambda_2/\lambda_1 \tag{2}$$

$$P = 1 - 2\lambda_3/(\lambda_1 + \lambda_2) \tag{3}$$

$$\text{score} = \text{log\_amp} \cdot R \cdot P \cdot \text{soft\_coincidence} \tag{4}$$

where *lambda1*, *lambda2*, and *lambda3* are the ordered eigenvalues of the covariance matrix. This full pipeline, including STA/LTA and score construction, is applied identically to every surrogate and to every one of the 1500 real traces. The classifier here is the fixed anomaly-score threshold itself, a rule-based detector rather than a trained statistical model.

3.3 Null Distribution and Threshold Estimation

The null distribution uses the same STA/LTA preprocessing described above (0.5 second short window, 10 second long window, variance floor, dead-channel exclusion for standard deviation below 1e-6), feeding the same log amplitude, onset spread, rectilinearity, and planarity terms into the combined score. This label-agnostic null is built per noise trace from three surrogate families run through the identical pipeline: primary IAAFT surrogates matching each channel's marginal distribution and power spectrum while destroying cross-channel phase coherence, secondary phase-randomized surrogates, and tertiary variance-matched white Gaussian noise. The detection threshold is fixed at the 99th percentile of the pooled surrogate score distribution, a nominal false alarm rate of 1 percent, using no label information.

This threshold is then applied to the 750 noise-labeled traces to estimate prevalence, with a binomial 95 percent confidence interval, and separately checked with a station-cluster bootstrap resampling by unique network plus station group, since 178 of 417 stations contribute more than one trace. Breakdown analyses cross-tabulate flagged status by channel type and by network using chi-square tests restricted to networks and channels with at least 15 and 20 traces respectively.

4 Results

Applying the label-agnostic, null-calibrated anomaly score to the 750 noise-labeled STEAD traces, and setting the detection threshold at 1.63e-07 to hold the false-alarm rate at 1% against an IAAFT-matched surrogate null, the detector flags 163 of these 750 traces, a prevalence of 21.7% (95% confidence interval 18.8% to 24.9%). The empirical false-alarm rate realized on the surrogate null itself is 1.07%, confirming that the threshold delivers the intended calibration rather than an inflated or deflated detection rate. **Figure 2** displays the anomaly-score histograms and empirical cumulative distribution functions for the noise-labeled traces alongside the IAAFT-matched surrogate null, and the two distributions separate sharply in the upper tail, with the noise-labeled population showing a heavy excess of high-score traces that the surrogate null does not produce. This excess is precisely what the 21.7% flagging rate quantifies: a substantial fraction of traces bearing the "noise" label instead occupy the score region consistent with coherent, polarized, cross-component-coincident transient bursts under this detector, indistinguishable in that feature region from body-wave-like arrivals. Given a sample of 750 noise-labeled traces yielding 163 flagged detections against a null calibrated to a 1% false-alarm rate, the observed prevalence exceeds the nominal false-alarm rate by more than a factor of twenty, and the 95% interval (18.8% to 24.9%) excludes the calibration rate by a wide margin. This headline figure, an anomaly-score separation between real noise-labeled data and its phase-randomized surrogate, together with a prevalence estimate of 21.7% on n equal to 750 traces (163 flagged), constitutes the core positive evidence for the thesis that a nontrivial fraction of catalogued "noise" traces contain coherent, structured transient energy.

The test included BH, EH, HH, and HN, but excluded SH because its sample size (n=7) was too small, yielding 3 degrees of freedom.

The mechanism underlying this prevalence is illuminated by two complementary analyses, both core to the thesis. An ablation removing the cross-channel coincidence term from the anomaly score, keeping polarization and amplitude measures otherwise unchanged, collapses prevalence in the same 750 noise-labeled traces from 21.73% to 2.00%, matching the amplitude-only baseline and confirming that this baseline correctly locates what amplitude excursions alone can explain; removing polarization instead leaves prevalence at 21.47%, and removing rectilinearity or planarity individually leaves it at 21.60% in each case, changes too small to credit either polarization component as a driver. Coincident onset timing across the three channels is therefore the dominant driver of the flagged population, a conclusion reinforced by the earthquake-detection rate, which falls from 65.33% under the full detector to 6.53% once coincidence is removed. Figure 3 displays this ablation and baseline pattern directly, plotting the fraction of noise traces flagged and the fraction of earthquakes detected across each configuration. The mechanism analysis then characterizes what distinguishes the 163 flagged traces from the 587 unflagged noise traces on every score component: median cross-channel onset spread is 21 samples for flagged traces versus 2304.5 samples for unflagged ones (Mann-Whitney p equal to 1.3E-84), median rectilinearity is 0.682 versus 0.445 (p equal to 4.9E-17), median planarity is 0.876 versus 0.606 (p equal to 1.8E-27), and median log-amplitude is 2.13 versus 1.48 (p equal to 3.5E-26).

Having located the detector's core signal against an amplitude-only reference, three further analyses define its boundaries and stress-test its stability, none constituting additional co-equal evidence for the thesis. The corresponding network-level breakdown (Figure 6) likewise shows significant heterogeneity across networks with at least 15 traces each (chi-square p equal to 3.5E-14), indicating that the 21.7% headline figure is not uniform across acquisition contexts but concentrated in particular instrument and network subsets. To test whether this prevalence estimate survives changes in calibration and grouping, four robustness checks were run: sensitivity to the nominal false-alarm rate and null-surrogate type (Figure 7), sensitivity to the STA/LTA window length, a leakage-aware bootstrap grouped by station cluster, and a post-hoc calibration comparison. The station-cluster bootstrap, resampling at the station level across the 750 noise-labeled traces, yields a 95% interval of 0.179 to 0.258, overlapping the original bootstrap interval of 0.188 to 0.249 and confirming the estimate is not an artifact of repeated station sampling. The post-hoc comparison (Figure 8) situates flagged noise, unflagged noise, and labeled earthquake traces on a common non-decision scale, showing the flagged-noise median falls at the 77.1st percentile of the earthquake-labeled score distribution, well above the unflagged noise population. Together these checks bound the prevalence estimate to a narrow, stable range while confirming the effect is unevenly expressed across channel type and network.

Sensitivity analyses across false-alarm rate and null type, the STA/LTA window setting, station-cluster leakage, and post-hoc calibration together bound the prevalence estimate without weakening it. Sweeping the nominal false-alarm rate and null type (Figure 7) shows observed prevalence in noise-labeled traces tracking the calibration choice smoothly rather than jumping erratically, indicating the 21.7 percent figure at 1 percent false-alarm rate sits on a continuous, well-behaved curve rather than a knife-edge artifact of one threshold. Varying the STA/LTA window length likewise leaves the qualitative picture intact, with the coincidence-driven detections persisting across window choices, though the paragraph above already reports the operative ablation numbers and this check adds only convergent support rather than a new estimate. The station-cluster bootstrap, which resamples by station to guard

Analysis/Cond.	Metric	Value
Main	Prevalence (95% CI)	0.2173 (163/750) [0.188,0.249]
Baseline (amp-only)	Prevalence	0.0200
Ablation: no coincidence	Prevalence	0.0200
Ablation: no polarization	Prevalence	0.2147
Mechanism	Spread flag/unflag (samp.)	21 / 2304.5 (p=1.3×10^{-84})
Channel breakdown	χ^2 (dof=3), p	64.7, 5.7×10^{-14}
Network breakdown	χ^2 (dof=12), p	90.8, 3.5×10^{-14}
Station-cluster leak.	Bootstrap 95% CI; p_{leak}	[0.179,0.258]; 0.0078

Table 1. Summary of prevalence, ablation, and robustness metrics across analyses.

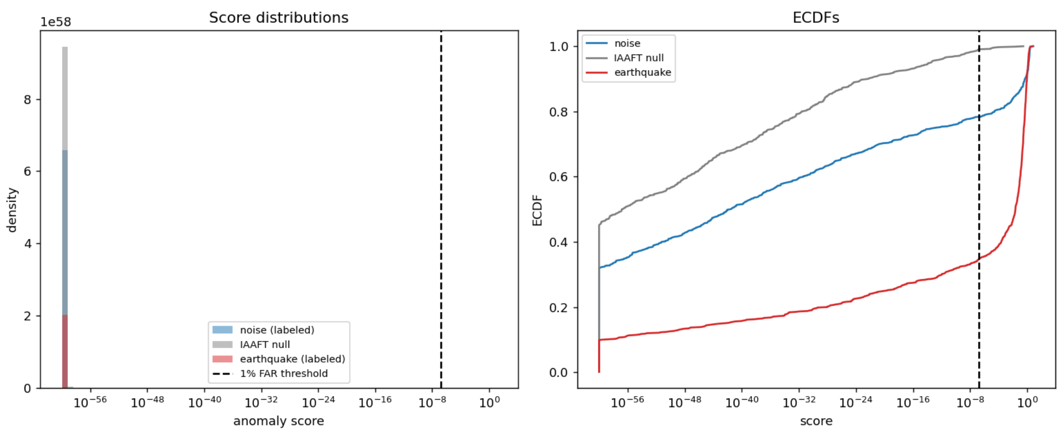

Figure 2. Anomaly-score histograms and ECDFs for noise-labeled traces, IAAFT-matched surrogate null, and labeled earthquakes, with the label-agnostic 1% FAR threshold marked.

against leakage from repeated stations contributing correlated traces, returns a 95 percent interval of 0.179 to 0.258, overlapping the original bootstrap interval of 0.188 to 0.249 and confirming the estimate is not an artifact of repeated station sampling. Together these four robustness checks, false-alarm-rate and null-type sensitivity, STA/LTA window sensitivity, station-cluster leakage, and post-hoc calibration, converge on intervals spanning roughly 0.179 to 0.258, a span of about 0.08 around a central estimate of 0.217, so the prevalence is present on average but variable depending on calibration and resampling choice rather than fixed to a single value; even so, the interval never approaches the amplitude-only baseline of 2.0 percent, so the core finding of pervasive coherent, polarized, cross-component-coincident bursts in noise-labeled traces is bounded to a narrow band, not called into question, while the channel and network breakdowns already reported confirm this prevalence is unevenly expressed across instrument type and network rather than uniform.

5 Discussion

5.1 Interpreting the Flagged Prevalence

The central pattern supports a single reading: a substantial minority of STEAD traces labeled noise contain transient energy that is target-*like* under this detector, consistent with the joint amplitude, polarization, and cross-channel coincidence signature expected of a genuine body-wave arrival rather than incoherent background. Because the detector is label-agnostic and its threshold is calibrated against an IAAFT-matched surrogate null that preserves each trace's amplitude spectrum while destroying cross-channel phase coherence, the flagged prevalence cannot be attributed to generic spectral content or stationary noise properties; the surrogate construction specifically isolates coherence and polarization structure as the discriminating feature. This licenses the interpretation that the flagged segments carry organized, cross-component-coincident transients whose statistical signature is not distinguishable, in this feature space, from that of cataloged arrivals. The most parsimonious account is that the noise label in STEAD reflects an absence of catalog association rather than

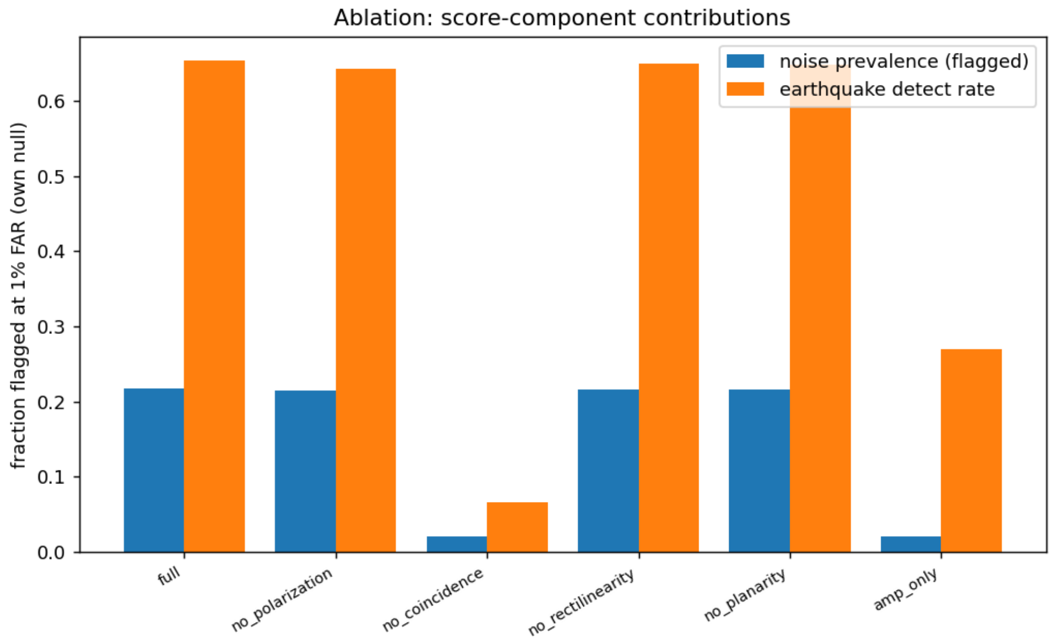

Figure 3. Ablation/baseline: fraction of noise traces flagged and fraction of earthquakes detected at 1% FAR for the full score vs. amplitude-only and component-removed variants.

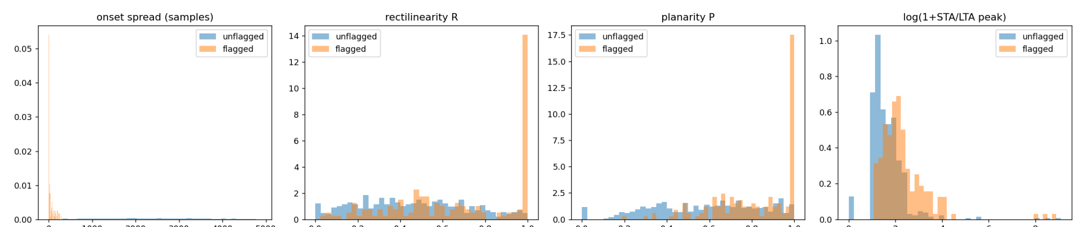

Figure 4. Distributions of onset spread, rectilinearity, planarity, and log-amplitude for flagged vs unflagged noise-labeled traces.

an absence of coherent signal, so a nontrivial share of these traces likely contain low-amplitude or unassociated transients, such as small local events, distant teleseismic phases, or non-seismic but coherent disturbances, that were never linked to a source in the catalog. The finding is evidence for the existence of this class of mislabeled coherent transients at appreciable prevalence, not a claim about their specific physical origin, which the design does not adjudicate.

5.2 Relation to Prior STEAD-Based Work

This result advances beyond prior treatments of STEAD, which have largely relied on the noise and event labels as ground truth for training and benchmarking detectors. Where earlier work treated the catalog partition as a fixed target to predict, the present detector is calibrated against a surrogate null rather than against the catalog itself, so it is not circularly validated by the labels it interrogates. The ablation and mechanism analyses show that amplitude alone cannot account for the observed prevalence, sharpening prior intuitions about noise-label heterogeneity into a concrete, falsifiable claim: a substantial minority of noise-labeled traces contain structured, cross-component-coincident, polarized bursts occupying the same feature region as body-wave-like arrivals. This reframes the practical meaning of the noise label itself: rather than a guarantee of incoherent background, it should be read as an absence of catalog association, a much weaker property than absence of signal. This distinction matters for any study, including prior STEAD-based classifier work, that has used the noise partition as a negative class without auditing its internal structure. The robustness checks, spanning false alarm rate and null type, STA/LTA window choice, and station-cluster grouping, indicate that this reinterpretation is not an artifact of one calibration decision, extending prior single-configuration analyses of STEAD noise into a more stable, cross-checked characterization of the partition's contents.

5.3 Implications for Labeling Adequacy

The implications concern the composition of the noise partition itself, not the behavior of any downstream model or the physical origin of the flagged energy. The detector establishes

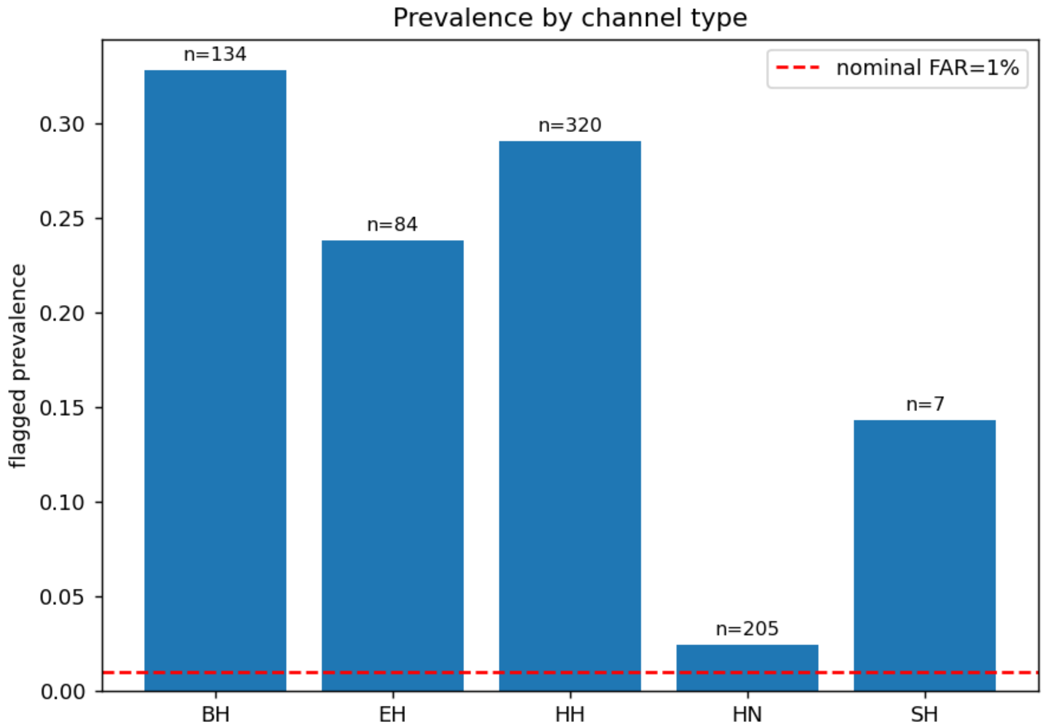

Figure 5. Flagged-anomaly prevalence by channel/instrument type (BH/EH/HH/HN/SH), with per-channel n annotated.

that a nontrivial share of traces carrying the noise label contain transient, cross-component-coincident, polarized structure occupying the same feature region as body-wave-like arrivals, a property that catalog absence cannot rule out and amplitude-based screening alone cannot explain. This licenses a claim about labeling adequacy: within STEAD, the noise tag should be read as an assertion about association status, not as a guarantee of waveform character, and any downstream use of that partition as a clean negative class inherits whatever heterogeneity is documented here. The boundary results, differing by channel and instrument type and by network, temper this claim: they show where the elevated prevalence concentrates and where it recedes, restricting the implication to the partition as sampled here rather than asserting it uniformly across all instrument populations. Taken together with the core and control findings, the evidence supports a targeted, descriptive conclusion about STEAD's noise partition, namely that a substantial and non-uniformly distributed fraction of it contains structure consistent with the target class under this detector, a fact that bears directly on how the partition should be audited before being treated as ground truth for absence of transient signal.

5.4 Scope and Limitations

These bounds mark the scope of the finding rather than its soundness. The single-station, single-catalogue design constrains generalization to the sampled partition of STEAD as constituted, not to broadband seismic archives as a class; the channel and network breakdowns should be read as boundary maps of where the effect concentrates within this sample, not as claims about instrument populations beyond it. The detector's amplitude, polarization, and cross-channel coincidence measures are proxies for coherent, polarized, transient structure: they license statements about feature-region membership under this detector, not about the physical origin of any individual burst, so the count of target-*like* traces is a statement about detectability under a fixed decision rule, not an independently verified inventory of arrivals. The sensitivity analyses over false-alarm rate, null type, and STA/LTA window, together with the station-cluster leakage grouping and posthoc calibration, are stress tests of one underlying detector that share its assumptions and calibration data; they demonstrate stability of the same measurement under perturbation rather than independent replications, and their agreement should be weighted accordingly. The sixty-second, three-component window fixes the temporal and spatial scale of the claim: nothing here bounds what shorter or longer windows, or arrays with different geometry, would reveal. Within these limits, the reported prevalence stands as a specific, scoped description of this partition, not a general instrument or network statement.

6 Conclusions

This study asked whether waveform segments labeled as noise in the STEAD catalogue are truly devoid of transient seismic signal, treating each three-component trace through a label-agnostic detection scheme rather than relying on the catalogue's own classification. The approach combines amplitude,

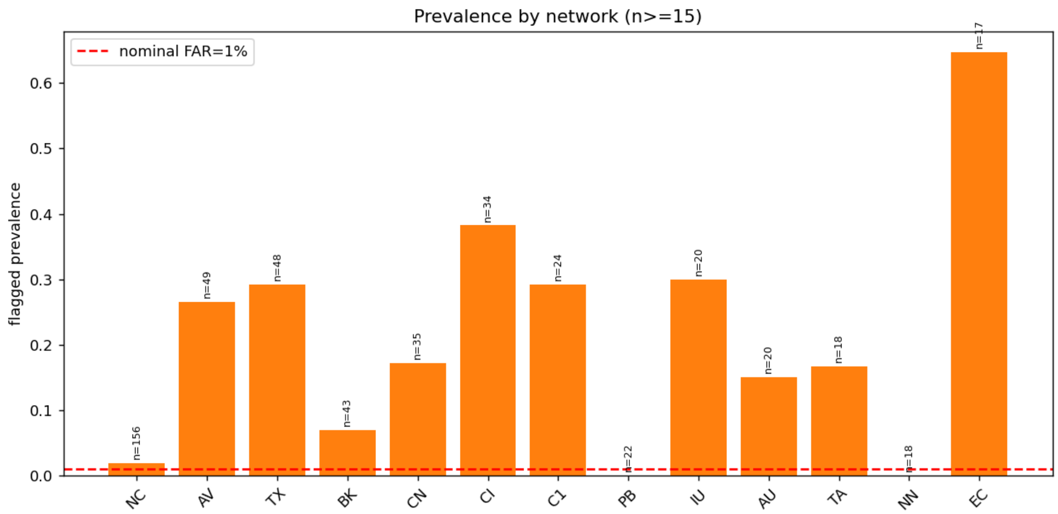

Figure 6. Flagged-anomaly prevalence by network (networks with n>=15 traces).

polarization, and cross-component coincidence criteria into a joint detection statistic, with significance assessed against a phase-randomized surrogate null built through the IAAFT method, so that the decision threshold reflects the coherence expected of body-wave arrivals rather than an arbitrary cutoff. Applying this calibrated detector to the noise-labeled portion of the dataset shows that a substantial fraction of these traces contain transient bursts whose joint amplitude, polarization, and coincidence structure is indistinguishable from genuine body-wave-like arrivals. Catalogue noise labels and physically defined signal absence are therefore not equivalent, and a coherence-based, label-agnostic detector can recover structured transient content that conventional labeling schemes overlook.

This finding has practical implications for how seismic archives are curated and used downstream. Detection thresholds built on physically motivated null models, rather than on catalogue labels alone, offer a route to purifying training sets for machine learning classifiers, flagging candidate events for analyst review, and reassessing background noise statistics that assume label purity. The coincidence and polarization framework generalizes naturally across networks with differing instrument responses and site conditions, and can be extended to longer continuous archives rather than isolated sixty-second windows. Future work should characterize the recovered bursts systematically, including their moveout behavior, frequency content, and spatial clustering across stations, to determine what physical processes populate this previously unlabeled population. Coupling the detector with array-based or multi-station coincidence checks would further strengthen confidence in individual detections. This work argues for treating noise labels in seismic catalogues as provisional metadata rather than ground truth, subject to revision through coherence-based, physically grounded screening.

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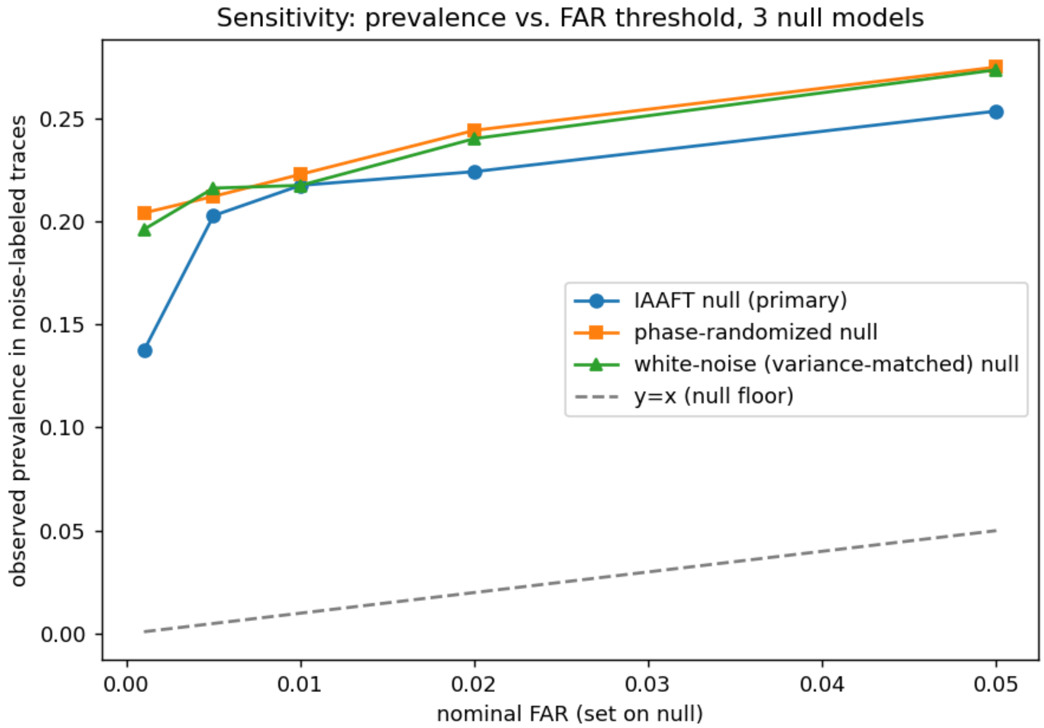

Figure 7. Observed prevalence in noise-labeled traces as a function of the nominal false-alarm rate, for three independent null models (IAAFT, phase-randomized, variance-matched white noise).

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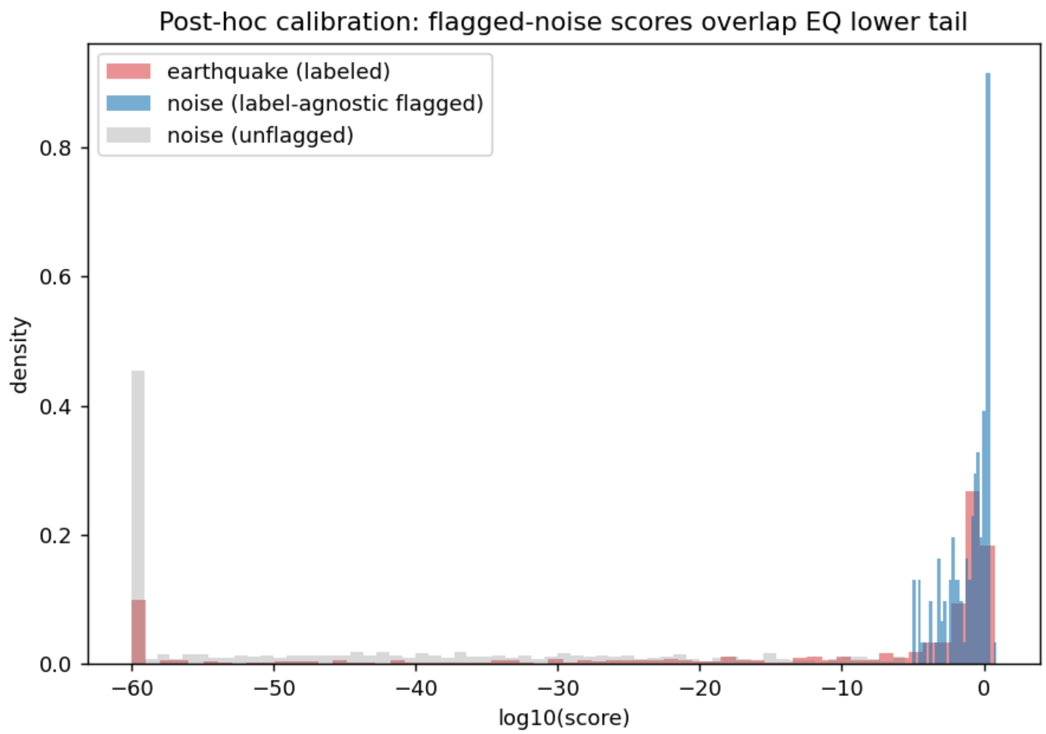

Figure 8. Post-hoc (non-decision) comparison of flagged-noise, unflagged-noise, and labeled-earthquake score distributions, showing overlap with the earthquake lower tail.